ADITYA KUMAR SINGH

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PROFILE

Motivated Computer Science and Business Systems undergraduate with hands-on experience in developing AI-powered applications using Python, FastAPI, Ollama, Retrieval-Augmented Generation (RAG), and Docker. Skilled in building locally hosted AI systems, voice assistants, semantic search applications, and multi-agent workflows. Seeking entry-level Software Engineering, AI/ML, or Generative AI opportunities to apply technical skills and build impactful solutions.

EDUCATION

B.Tech in Computer Science and Business Systems (CSBS) **August 2022 – July 2026**
Techno India University, West Bengal **CGPA: 7.54/10**

Higher Secondary School **August 2019 – July 2021**
A. A. M. Children's Academy, Katihar, Bihar **Percentage: 65.8%**

Secondary School **August 2017 – May 2019**
R. K. M. Vidya Mandir, Katihar, Bihar **Percentage: 63.8%**

PROJECTS

AURA – Autonomous AI Voice Assistant **Python, FastAPI, Ollama, Gemma 4**

- Built a locally hosted AI voice assistant using Python, FastAPI, Ollama, and Gemma 4, supporting both voice and text-based interactions.
- Implemented speech-to-text (STT), text-to-speech (TTS), wake-word detection, system control, OCR-based screen reading, web search, file management, and conversational memory.
- Developed a real-time Tactical HUD for system telemetry, hardware monitoring, and agent interaction.

Nyaya AI Offline – AI-Powered Legal Assistant **Python, Ollama, RAG, Docker, n8n, Pinecone**

- Developed an AI-powered legal assistant for Indian legal research using Retrieval-Augmented Generation (RAG) and locally hosted large language models (LLMs).
- Built semantic search using vector embeddings and Pinecone to retrieve relevant legal documents and generate context-aware responses.
- Implemented a multi-agent AI workflow comprising a Legal Assistant, Lawyer, Opposing Counsel, and Judge to simulate legal analysis and reasoning.
- Containerized the application using Docker and automated workflows with n8n to streamline deployment and document-processing tasks.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL

AI & Machine Learning: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Ollama, Vector Embeddings, Semantic Search, Multi-Agent Systems

Frameworks & Tools: FastAPI, Docker, n8n, Pinecone, Git, GitHub

Web Technologies: HTML, CSS, JavaScript

Operating Systems: Windows, Linux

Other: REST APIs, OCR, Speech-to-Text (STT), Text-to-Speech (TTS)

CERTIFICATIONS

- **Cybersecurity Virtual Internship** – EduSkills, supported by Palo Alto Networks **May – July 2023**
- **Android Developer Virtual Internship** – EduSkills, supported by Google for Developers **Sep – Nov 2023**
- **Networking Virtual Internship** – EduSkills, supported by Juniper Networks **Jan – Mar 2024**